

UNIVERSITAS
GADJAH MADA

Unconstrained Growth, Constrained Growth & Decay

Problem Solving Methods and Modeling

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Unconstrained Growth

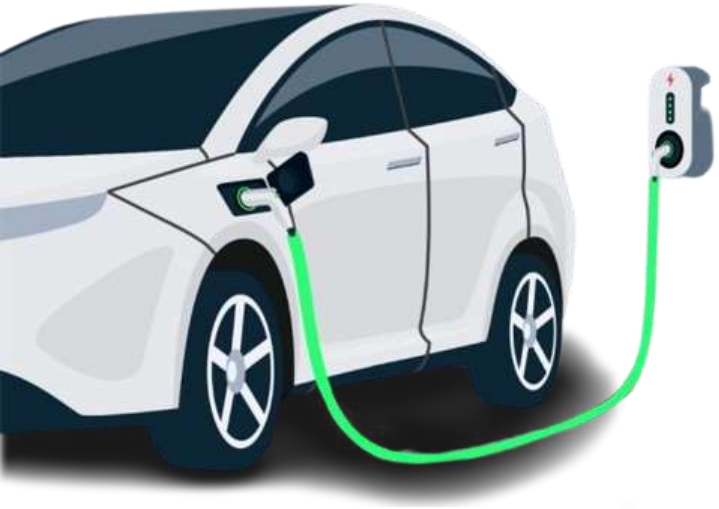
In December 2019, the first case of COVID-19 was detected in Wuhan, China. Since this virus was highly contagious and there were no interventions (such as lockdowns or vaccines), the number of cases initially grew exponentially. By early January 2020, the confirmed cases had reached approximately 1,000, with an estimated daily growth rate of 15% before preventive measures were implemented.

1. What will be the number of cases after 10 days if no interventions are taken?
2. On which day will the number of cases exceed 1 million?
3. Use Insight Maker to simulate the growth of COVID-19 cases over the first 60 days and observe the shape of the graph



Constrained Growth

A city has begun implementing charging stations for electric vehicles, leading to an increase in the number of users. Initially, there were only 1,000 active electric vehicles, but due to government incentives, the growth rate reached 15% per month. However, since the maximum infrastructure capacity can only support 50,000 electric vehicles, the growth eventually started to slow down.



1. If the initial growth rate is 15% per month, how many electric vehicles will there be after 12 months?
2. Use Insight Maker to simulate the growth of electric vehicles over 5 years and observe how the growth slows down.
3. If after 24 months the government increases the charging station capacity to accommodate 100,000 vehicles, how will the growth change?

Decay

An SSD (Solid State Drive) has a limited lifespan based on the number of read/write cycles. Over time, the SSD's ability to store data gradually declines exponentially, and after 5 years, its performance can drop by up to 70% of its original condition.



1. If the SSD performance degradation rate is 8% per year, what will be its performance after 5 years?
2. In which year will the SSD performance drop to 50% of its original condition?
3. Use Insight Maker to simulate how SSD performance decreases over time.

The answers to the questions above should be structured, as the score depends on the completeness of each item in the question.

THANK YOU